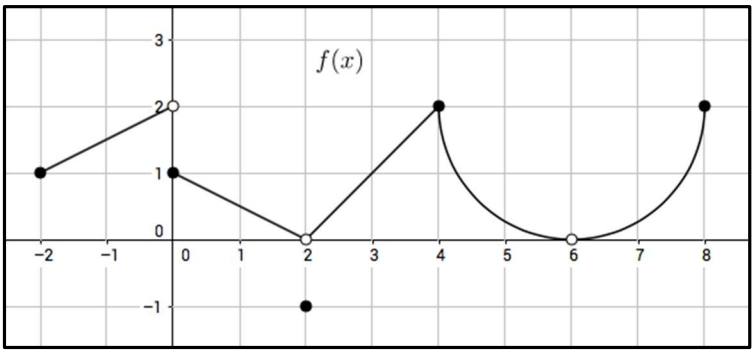
Worksheet 1-2 & 1-3 A: Limit Notation and Graphical Limits #1
AP Calculus Name _____

1. Translate the following into words:

$\lim_{x \to -3} g(x) = 6$ Translation: _____



2. Use the graph of $f(x)$ above to answer the following.

a. $\lim_{x \to 2} f(x) =$

b. $\lim_{x \to 0^-} f(x) =$

c. $f(0) =$

d. $\lim_{x \to 0} f(x) =$

e. $f(2) =$

f. $\lim_{x \to 1} f(x) =$

3. Describe the difference between the following two statements:

Statement I: $\lim_{x \to a} h(x) = c$

Statement II: $h(a) = c$

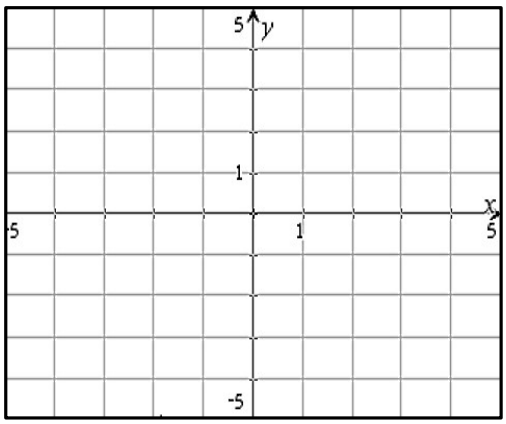
4. Sketch a graph with the following properties:

$\lim_{x \to 1} f(x) = 4$

$f(1)$ is undefined

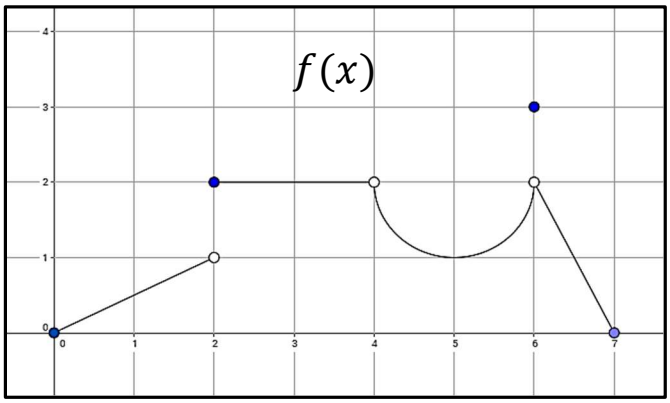
$\lim_{x \to -2^-} f(x) = 3$

$\lim_{x \to -2^+} f(x) = 1, \ f(-2) = 0$



Created by B Passwater

Directions: Use the graph below to answer questions 5 – 13.



5. $f(2) =$

6. $\lim_{x \to 2} f(x) =$

7. $\lim_{x \to 4} f(x) =$

8. $f(4) =$

9. $\lim_{x \to 2^+} f(x) =$

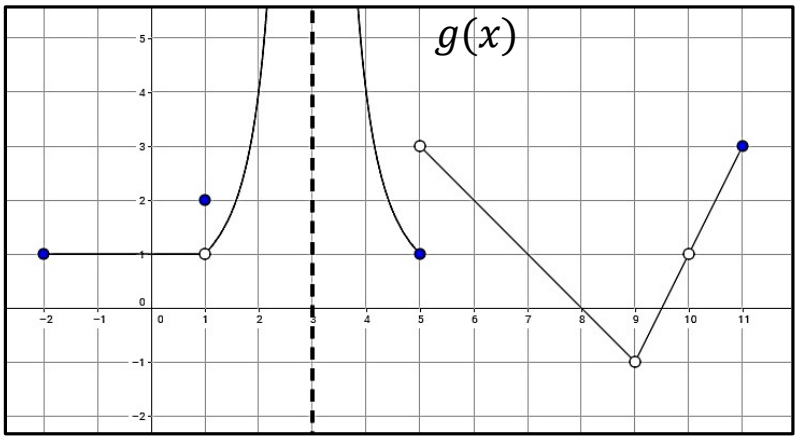
10. $\lim_{x \to 2^-} f(x) =$

11. $\lim_{x \to 6^-} f(x) =$

12. $\lim_{x \to 6} f(x) =$

13. $f(6) =$

Directions: Use the graph below to answer questions 14 – 22.



14. $\lim_{x \to 7} g(x) =$

15. $\lim_{x \to 1} g(x) =$

16. $g(1) =$

17. $g(3) =$

18. $\lim_{x \to 10} g(x) =$

19. $\lim_{x \to 3} g(x) =$

20. $\lim_{x \to 5^+} g(x) =$

21. $\lim_{x \to 5} g(x) =$

22. $\lim_{x \to 9} g(x) =$